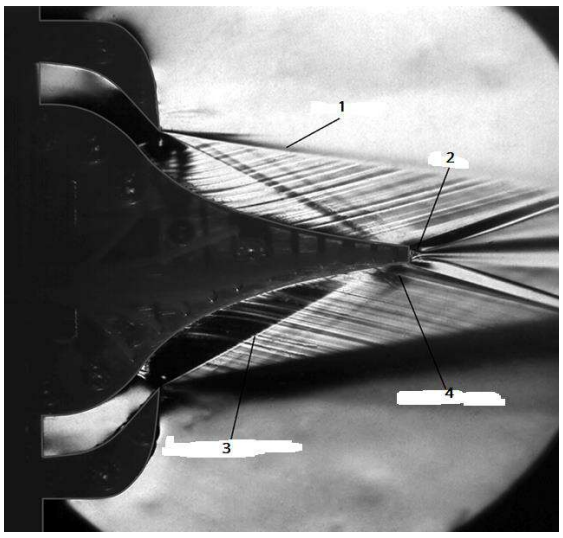
Gas Dynamics Written Test July 17th 2023 (MSAR B. Favini)

1. The schlieren photograph shows a steady two-dimensional supersonic flows for an aerospike nozzle. The expansion ratio (total pressure / external pressure) is equal to 25. Describe the flow field and identify the flow features labeled 1-2-3-4.



La fotografia Schlieren mostra un flusso supersonico bidimensionali stazionario per un ugello aerospike. Il rapporto di espansione (pressione totale / pressione esterna) è pari a 25. Descrivere il campo di flusso e identificare le caratteristiche del flusso etichettate 1-2-3-4.

(10/30)

2. A plane shock propagates inside a constant section duct. The air upstream of the shock is at rest with the pressure p=1 and the speed of sound a=1. The absolute Mach number downstream of the shock is M=1.87. The shock interacts with an abrupt reduction in area equal to 1.5. Determine the change in the shock intensity due to its interaction with the area change. Sketch the flowfield on the x-t and p-u planes.

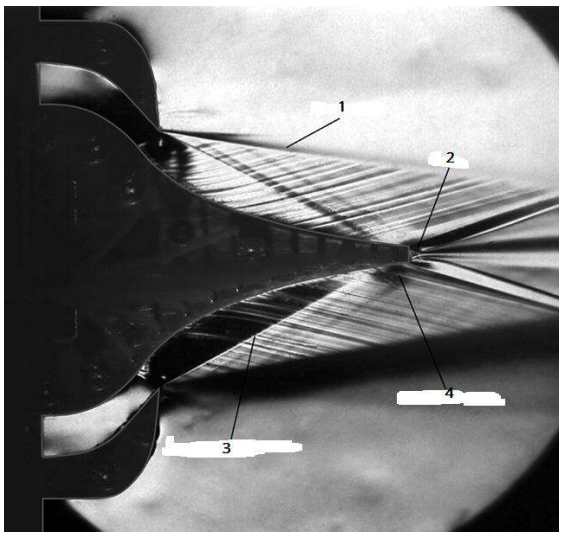


Un urto piano propaga all'interno di un condotto a sezione costante. L'aria a monte dell'urto è in condizione di quiete con la pressione p=1 e la velocità del suono a=1. Il numero di Mach assoluto a valle dell'urto M=1.87. L'urto incontra una brusca riduzione d'area pari a 1.5. Determinare la variazione dell'intensità dell'urto a causa della sua interazione con la variazione d'area. Disegna il flusso sui piani x-t e p-u.

(20/30)

Solutions Gas Dynamics Written Test July 17th 2023 (MSAR B. Favini)

1. The schlieren photograph shows a steady two-dimensional supersonic flows for an aerospike nozzle. The expansion ratio (total pressure / external pressure) is equal to 25. Describe the flow field and identify the flow features labeled 1-2-3-4.



The schlieren shows an under-expanded supersonic plane jet, free on the ambient side and confined on the plug side. The flow along the plug wall exhibits characteristic lines which means that it is not a simple wave flow.

The boundary between the jet and the ambient is a shear layer (1).

The truncated exit geometry of the plug generates a wake whose elements are: a Prandtl-Meyer expansion followed by a recirculating region and a recompression shock (2).

The Prandtl-Meyer expansion at the nozzle exit, where the flow enters the ambient, reflects at the jet boundary as a compression that coalesces in a shock (3). This almost straight shock reflect on the plug wall (4).